What is the impact of GDP and urban population in the US and its annual CO_2 production since 1960 and how can it be predicted for the next 5 years?

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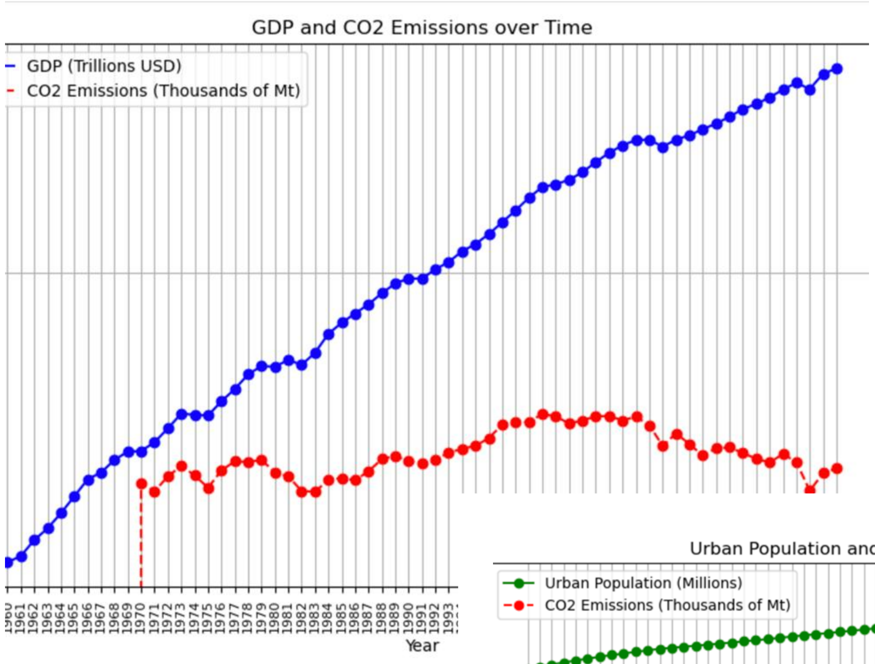
CMSE201-008

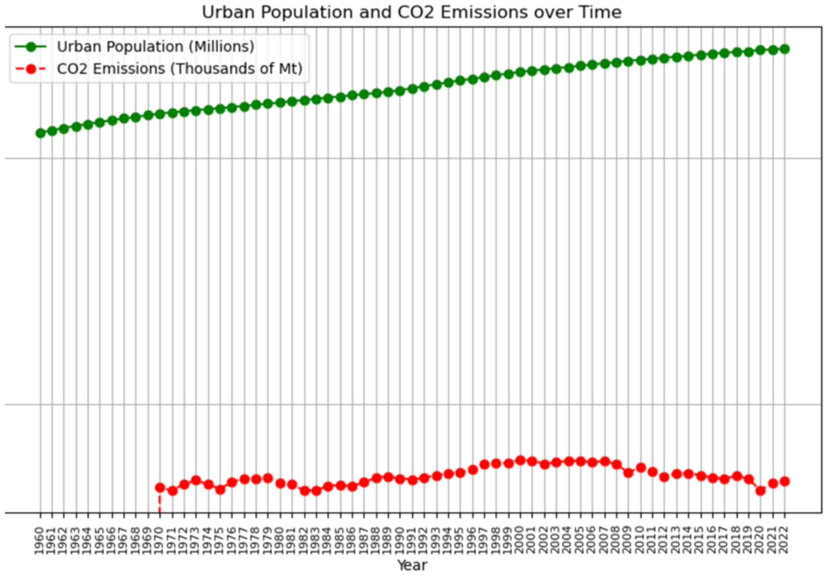
Impact

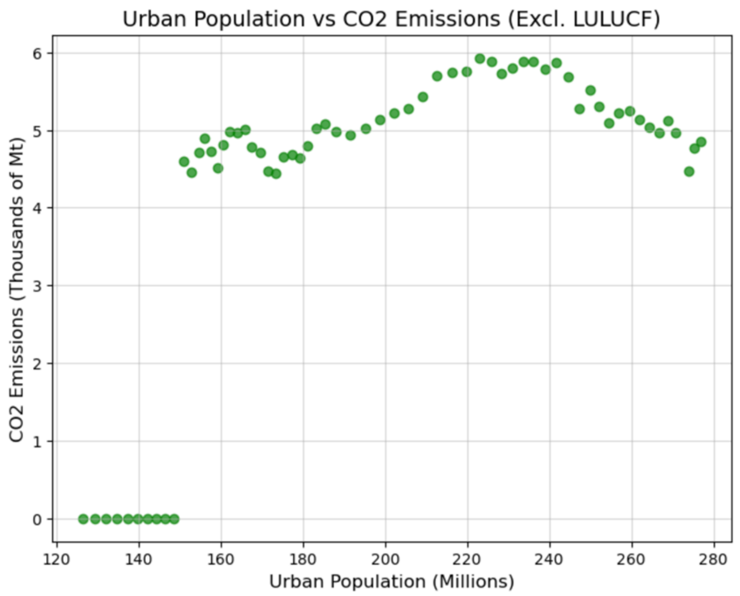
General positive correlation until early 2000's.

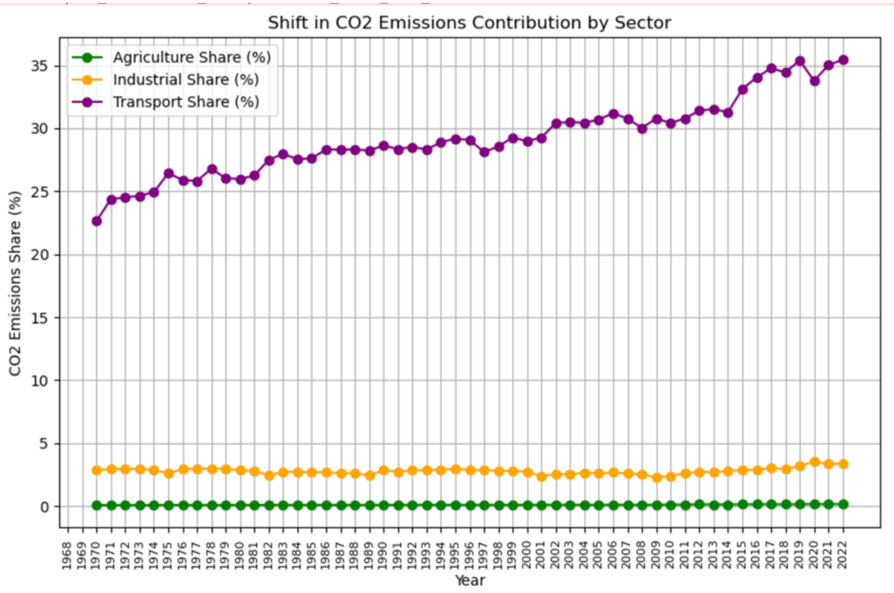
The 'dips' in the CO_2 emissions correspond to external factors: pandemic, recession.

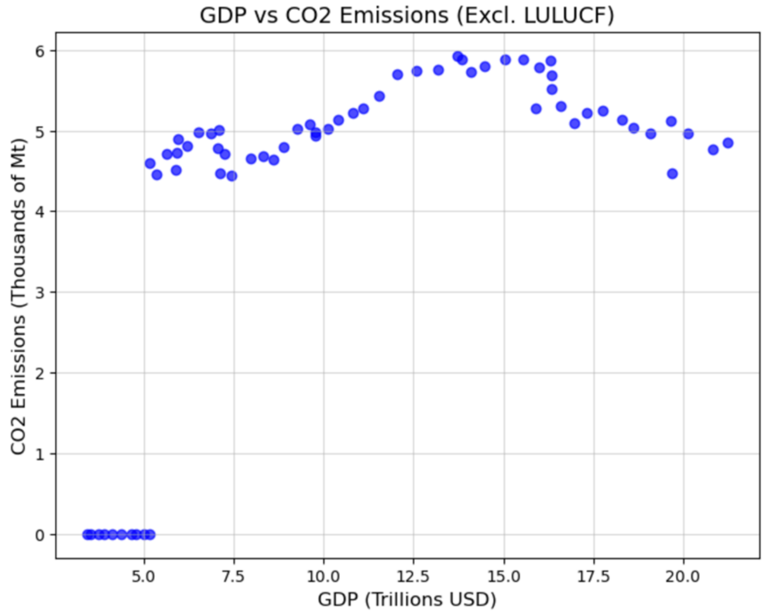
```
# Plot GDP vs CO2 emissions over time
plt.figure(figsize=(10, 6))
plt.plot(years, gdp / 1e12, label='GDP (Trillions USD)', marker='o', color='blue')
plt.plot(years, co2_total_excl_lulucf / 1e3, label='CO2 Emissions (Thousands of Mt)', marker='o', linestyle='--', color='red')
plt.xlabel('Year')
plt.ylabel('Values')
plt.title('GDP and CO2 Emissions over Time')
plt.xticks(rotation=90, fontsize=8)  # Rotate x-axis labels vertically and shrink font size
plt.legend()
plt.grid()
plt.show()

# Plot Urban Population vs CO2 emissions over time
plt.figure(figsize=(10, 6))
plt.plot(years, urban_population / 1e6, label='Urban Population (Millions)', marker='o', color='green')
plt.plot(years, co2_total_excl_lulucf / 1e3, label='CO2 Emissions (Thousands of Mt)', marker='o', linestyle='--', color='red')
plt.xlabel('Year')
plt.ylabel('Values')
plt.title('Urban Population and CO2 Emissions over Time')
plt.xticks(rotation=90, fontsize=8)  # Rotate x-axis labels vertically and shrink font size
plt.legend()
plt.grid()
plt.show()
```












```python
agriculture_share = co2_agriculture / co2_total_excl_lulucf
industrial_share = co2_industrial / co2_total_excl_lulucf
transport_share = co2_transport / co2_total_excl_lulucf

plt.figure(figsize=(10, 6))
plt.plot(years, agriculture_share * 100, label='Agriculture Share (%)', marker='o', color='green')
plt.plot(years, industrial_share * 100, label='Industrial Share (%)', marker='o', color='orange')
plt.plot(years, transport_share * 100, label='Transport Share (%)', marker='o', color='purple')
plt.xlabel('Year')
plt.ylabel('CO2 Emissions Share (%)')
plt.title('Shift in CO2 Emissions Contribution by Sector')
plt.xticks(rotation=90, fontsize=8)  # Rotate x-axis labels vertically and shrink font size
plt.legend()
plt.grid()
plt.show()
```

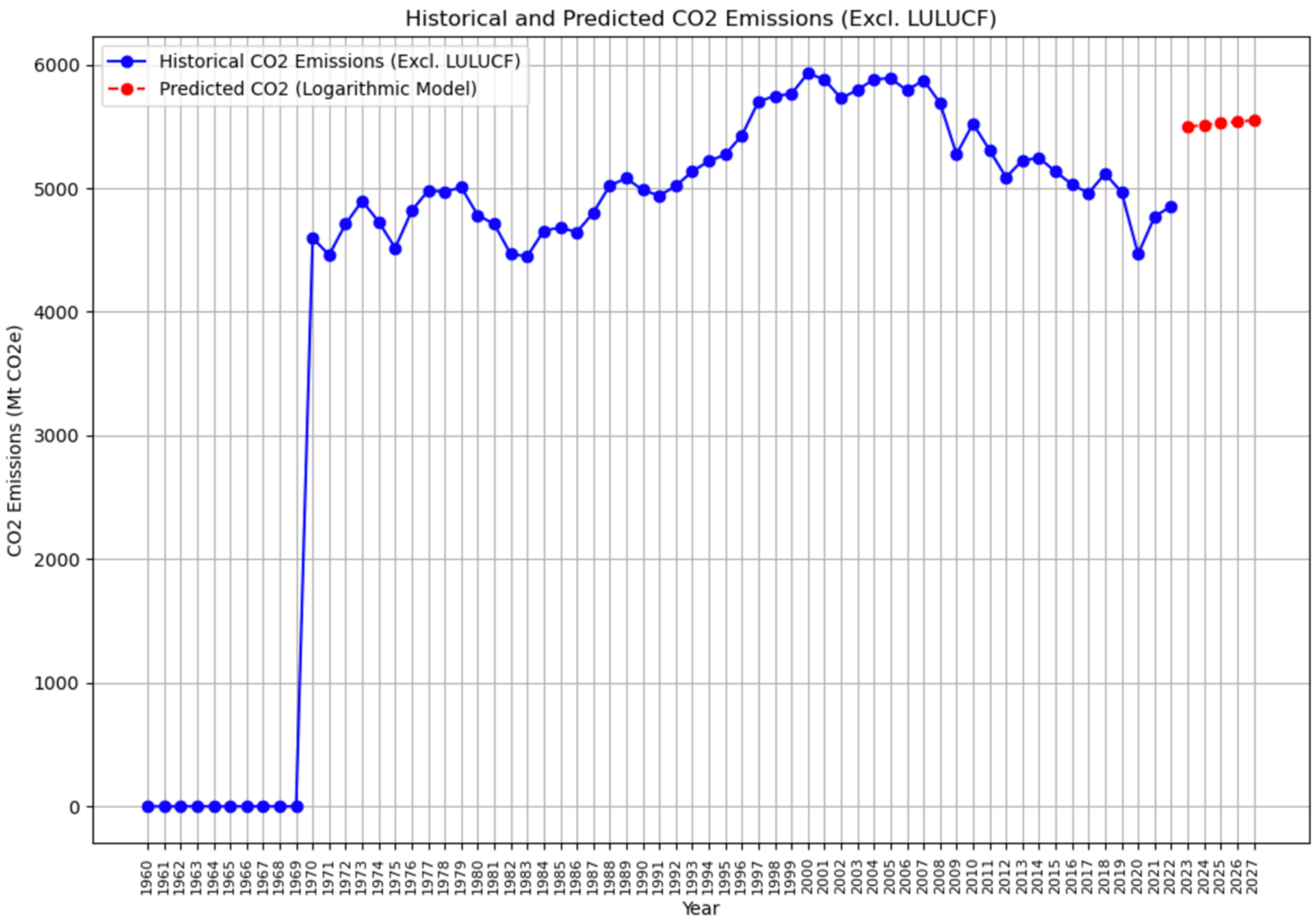
Impact

Correlation between GDP and CO_2 emissions: 0.6115

Correlation between Urban Population and CO_2 emissions: 0.6480

Modelling

Logarithmic model function is good for data that exhibits nonlinear relationships, diminishing returns, or extreme variation.




```python
def log_model_func(x, a, b, c):
    return a * np.log(x[0]) + b * np.sqrt(x[1]) + c
```

```python
# Perform curve fitting for logarithmic model
popt_log, pcov_log = curve_fit(log_model_func, x_data, y_data)

# Predict CO2 emissions for the next 5 years
future_years = list(map(str, range(end_year + 1, end_year + 6)))
gdp_future = np.linspace(gdp[-1], gdp[-1] * 1.05, 5)  # Assuming a 5% growth over 5 years
urban_future = np.linspace(urban_population[-1], urban_population[-1] * 1.02, 5)  # Assuming 2% growth
data_future = np.array([gdp_future, urban_future])

# Predictions for both models
predicted_co2_log = log_model_func(data_future, *popt_log)
```

Conclusion

- There is a weak correlation between GDP/Urban population and the CO_2 emissions.
- Inaccuracy in model suggests that there are other factors that affect emissions.
- More complex modelling tools like a regression or machine learning model would yield better results.